



AI Driven SQA: Manual & Automation Testing

Assignment 01

Module: 01

**Topic: Introduction to AI-Driven Software
Quality Assurance (SQA).**

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Q1. What is Software Quality Assurance (SQA)?

Answer: Software Quality Assurance (SQA) is a planned and organized way of activities that make sure software meets marked quality standards. It pinpoints the processes used to develop software rather than only checking the final product. The main goal of SQA is to prevent defects before they occur in live deployment and ensure that the software satisfies customer requirements. SQA is applied throughout the entire Software Development Life Cycle (SDLC). It goes through requirement gathering to production support.

Quality and usability is very important in software development. Because it helps build customer trust and protects an organization's reputation. Major failure or minor failure in software live deployment can cause users to lose confidence in a product. In case of high quality software helps to protect revenue by reducing downtime and system failures. In industries as banking and healthcare quality is essential for maintaining security and complying with regulations. Good software quality improves user satisfaction and retention. While poor quality and overly complicated software may cause users to stop using the application. So finding and fixing bugs early reduces long term development and maintenance costs.

So SQA ensures that software is developed using effective processes and required quality standards. It increases customer satisfaction, improves reliability, enhances security and reduces overall costs.

Q2. Differentiate between the following:

Answer:

QA vs QC

Aspect	Quality Assurance (QA)	Quality Control (QC)
Focus	Process-oriented	Product-oriented
Goal	Prevent defects	Identify defects
Approach	Proactive	Reactive
Activities	Audits, standards, process reviews	Inspection, reviews of the product
Responsibility	Entire team	QC / QA team

QC vs Testing

Aspect	Quality Control (QC)	Testing
Focus	Product-oriented	Execution-oriented
Goal	Identify defects	Find bugs & verify behavior
Approach	Reactive	Reactive
Activities	Inspection, reviews of the product	Test case execution and exploratory testing
Responsibility	QC / QA team	Testers / Automation Engineers

QA vs Testing

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Q3. How is AI changing Software Testing?

Answer: Artificial Intelligence (AI) is transforming software testing by making testing activities faster, smarter and more efficient. Unlike manual testing that follow fixed scripts. AI testing can understand intent, adapt to changes and learn from previous test executions history.

AI is now used from start to finish in Software Development Life Cycle (SDLC). It assists in requirement analysis, test case generation, code review, test execution and defect triage. This allow QA teams to reduce manual effort and improve productivity. One important advancement is self-healing automation. When a user interface changes and a test locator breaks due to UI changes. AI can automatically identify the new element and update the test and reducing maintenance work. It also provides intelligent reporting by summarizing failures, grouping similar defects and helping identify main cause instead of only showing pass/fail logs. This help QA team to analyze issues more effectively.

AI is also used to generate test cases, create test data, assist in test execution, support bug reporting, generate automation scripts and help with documentation. It improve efficiency, reducing manual effort, supporting smarter automation and helping QA team deliver higher-quality software more quickly.

Q4. Explain the responsibilities of each role.

QA Engineer: A Quality Assurance (QA) Engineer is responsible for ensuring that software development follows client defined requirement, quality standards and processes. The main objective of a QA Engineer is to prevent defects before they occur after launching and improve the overall development process.

Daily activity

- Review requirements for clarity.
- Participate in requirement and design reviews.
- Create and maintain test plans and test cases.
- Ensure quality standards and processes are followed.
- Monitor defects, security issue and quality metrics.
- Collaborate with DevOps, testers and project lead.
- Monitor development processes.

Software Test Engineer: A Software Test Engineer is responsible for verifying that software is working according to approved requirements and is free from critical defects before release.

Daily activity

- Design and execute test cases.
- Perform functional testing and defect reporting.
- Conduct retesting and regression testing.
- Perform integration and API testing.
- Conduct automation, performance, security Testing
- Report and track defects.
- Verify bug fixes.
- Prepare test execution reports.

AI Quality Engineer: An AI Quality Engineer ensures the quality, accuracy, reliability and security of AI systems. Besides standard QA task AI QA validate AI generated responses and test AI behavior. They also check for hallucinations, context understanding, prompt interpretation and security issues.

Daily Activities

- Test AI generated responses.
- Check response accuracy.
- Identify hallucinate information.
- Validate context understanding.
- Verify prompt understanding.
- Perform security testing on AI systems.
- Assist with bug reporting, automation script generation and documentation.
- Collaborate with DevOps and project managers.

Q5. Draw the career roadmap of a QA professional.

Answer: Roadmap flow from Junior QA Engineer to Head of AI Quality. Show in (Fig 5.1).

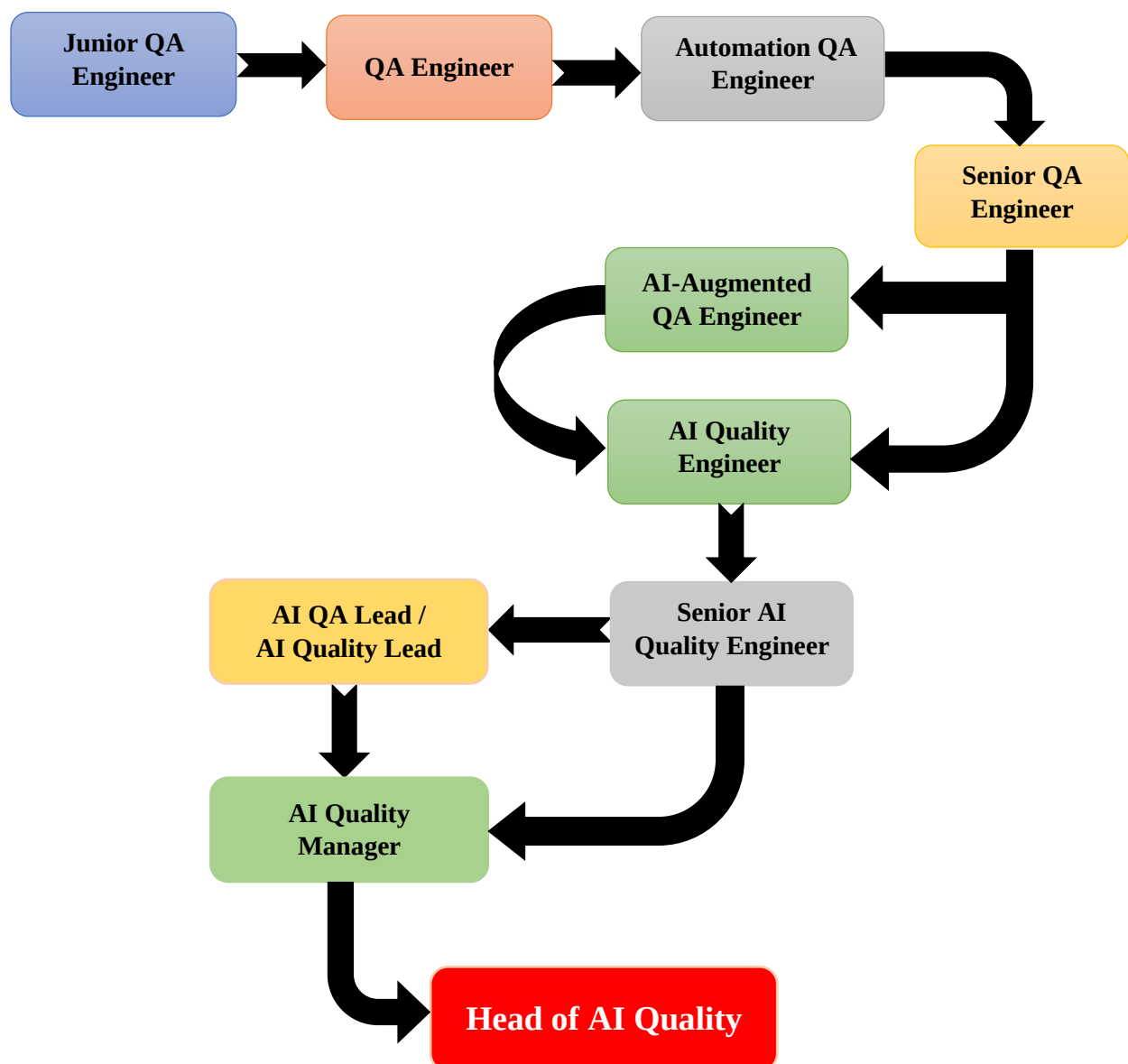


Fig 3.1: Career roadmap of a QA professional.